Amen Woldesenbet Brhanemesqel

Software Engineer · Research Assistant

291, Daehak-ro, Yuseong-gu, Daejeon, 34141, Rep. of KOREA

(+82) 10-3315-5536 | amita85412525@kaist.ac.kr | amenwolde.github.io | amenwolde | Amen Woldesenbet

Education

KAIST (Korea Advanced Institute of Science and Technology) — *Daejeon, South Korea*

B.S. in Computer Science and Brain and Cognitive Sciences, with a Full-ride Scholarship — *Aug. 2022 - Present*

- **Relevant Coursework:** Introduction to Design, Requirements Engineering for Smart Environments, Biology of Neurons, Neural Circuits for Cognition, Laboratory on Human Brain, Psychological and Behavioral Sciences.

AAiT (Addis Ababa Institute of Technology) — *Addis Ababa, Ethiopia*

AddisCoder Coding Camp (www.addiscoder.com) — *Jul. 2019 - Aug. 2019*

Experience

Interactive Computing Lab (Prof. Uichin Lee), KAIST — *Daejeon, South Korea*

Undergraduate Researcher — *Jul. 2025 - Current*

- Built the Android app system for a Wizard-of-Oz mobile automation agent used in the paper's user study with older adults, with an in-app floating chat overlay and a FastAPI backend with a wizard-side console.
- Quantitatively analyzed smartphone interaction logs (delegation versus intervention, interaction rate, active interaction span).
- Co-authored the IMWUT paper *Speak, Watch, Verify: A Mental Model of Older Adults' First-Use Interactions with Mobile Automation Agents*; under major revision, currently working on the resubmission.
- Prototyped a RAG-based Q&A assistant for older adults on YouTube as a preliminary study that informed the Wizard-of-Oz study.

Make Lab (Prof. Andrea Bianchi), KAIST — *Daejeon, South Korea*

Undergraduate Researcher — *Jan. 2026 - Current*

- Building a Unity-based virtual reality system for spatial audio navigation in situations where vision is unavailable.
- Designing VR study scenarios (path to exit, obstacle search) to test how far audio alone can support navigation, to inform navigation support for visually impaired people.
- Previously built a node-based image generation interface where each visual component is a node with parent-child dependencies on a radial canvas, so edits to a child recursively propagate up the hierarchy.
- The system composes images pairwise through the node graph (two at a time), letting users build scenes from more reference images than the underlying model normally supports while keeping identity drift under control.

Mental Health Accessibility Project, Batoners (CEO Lee Ji-su) — *Daejeon, South Korea*

UI/UX Design Intern — *Dec. 2024 - Feb. 2025*

- Redesigned an existing mental-health app for blind and deaf users, with haptic feedback, text-to-speech output, and visual aids.
- Wrote accessibility documentation for the development team.

Wormery Lab (Prof. Netta Cohen), University of Leeds — *Leeds, United Kingdom*

Research Intern — *Jul. 2026 - Aug. 2026*

- Studied whether a neural circuit that loses part of its structure to injury can recover its timing using only the connections it still has, motivated by recovery after spinal cord injury.
- Worked in a dynamical-systems framework, characterising limit cycles, phase-response curves and phase locking in pulse-coupled FitzHugh–Nagumo oscillators.
- Found that, within the model, surviving connections alone restored the injured unit's firing rhythm, and discussed the model's limits for recovery after spinal cord injury.

Brain Repair & Organoid Intel Lab (Prof. Minee Choi), KAIST — *Daejeon, South Korea*

Research Assistant — *Mar. 2024 - Dec. 2024*

- Conducted research on induced pluripotent stem cells (iPSCs) and their applications in personalized drug discovery.
- Studied the reprogramming process of somatic cells into iPSCs using Yamanaka factors and analyzed methods for differentiating iPSCs into cortical neurons.

Manuscripts

Speak, Watch, Verify: A Mental Model of Older Adults' First-Use Interactions with Mobile Automation Agents

IMWUT (PACM on Interactive, Mobile, Wearable and Ubiquitous Technologies)

Fourth Author | Major Revisions Decision, Revision in Progress — *2025 - Current*

- Wizard-of-Oz study with 30 older adults (60+) completing 20 everyday smartphone tasks with a voice-driven agent; proposes the Initiation–Intervention–Completion (IIC) model of user breakdowns.

Skills

Programming	Python, Kotlin, JavaScript, C/C++, SQL, HTML/CSS
Frameworks & Tools	Unity, React, Flutter, FastAPI, Docker, Git
ML & Data	RAG pipelines, ChromaDB, vision-language model prompting

Projects

ClothCall: Discreet Clothing Inspection for Blind Adults

Addis Ababa, Ethiopia & Daejeon, South Korea

Lead Developer & Researcher — *Mar. 2026 - Current*

- Ran a formative study with five blind participants in Ethiopia, in Amharic — narrowed scope to stain and fading detection, and surfaced a preference for a trusted person's judgment over AI analysis alone.
- Led development of the Android app in Kotlin and Jetpack Compose, using a vision-language model.
- Built a trusted-person calibration: a caregiver sets a reference photo and tolerance level, and the app refers to them by name instead of giving a generic AI output.
- Designed a phone-call interaction so the user holds the phone to their ear and hears the result privately.
- Co-authoring a paper on the project's design contributions.

Modelling Reorganisation and Recovery of Neural Circuits After Injury

University of Leeds, School of Computer Science

Research Intern — *Jul. 2026 - Aug. 2026*

- Built and validated a staged Python model, from a linear control oscillator to a ten-unit neuron chain.
- Chose timesteps by self-convergence and checked the code against the control system's analytic solution.
- Modelled injury as a cut connection and as a reduced drive current, then compared recovery in each.
- Internship report.

Wizard-of-Oz Mobile Automation Agent for Older Adults

IC Lab, KAIST, Daejeon, South Korea

Android & Backend Developer — *Oct. 2025 - Mar. 2026*

- Built the Android app system for a Wizard-of-Oz agent: a floating chat overlay that gives older adults voice and text guidance and structured choices on screen.
- Built a wizard console on a FastAPI backend, so a human operator guides users in real time by picking steps and choice templates instead of typing freely.
- Used the system to run the user study underlying the IMWUT manuscript.
- GitHub repo.

RAG-Based Q&A Assistant for Older Adults on YouTube

IC Lab, KAIST, Daejeon, South Korea

Developer — *Jul. 2025 - Aug. 2025*

- Prototyped a RAG-based Q&A assistant that gives older adults in-app help on YouTube, combining semantic FAQ retrieval with a small-LLM fallback and a chat overlay built as a Chrome extension.
- Used MiniLM embeddings and ChromaDB over 30 curated FAQs, with FLAN-T5-small as the fallback, served through FastAPI.
- Evaluated response time of FAQ retrieval versus the LLM fallback.
- Demo | GitHub repo.

GrabNGo Mobile App (Introduction to Software Engineering)

Daejeon, South Korea

Front-end Developer — *Sep. 2024 - Dec. 2024*

- Built the mobile app front end for a campus convenience store checkout system where customers walk in, shop, and walk out without a check-out line.
- Implemented registration, sign-in with password recovery, a welcome screen after face authentication, a live virtual cart, and the payment screen at walkout.
- Paired with the backend team to debug authentication and cart tracking, and tested usability.
- Demo.